

Rahmi Ramadhani Umar

Associate Professor of Mathematics Education · Universitas Potensi Utama, Medan
JESH Postdoctoral Fellow · Johannes Kepler University Linz, Austria

27

Documents, Scopus

9

h-index, Scopus

119

Documents, Scholar

31

h-index, Scholar

6

Patents and IP

PLATE I WOVEN MEASURES A Rasch response matrix: warp = item, weft = learner, gold = correct answer drawn with $P = 1/(1+e^{-(\theta - b)})$

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🔍 Google Scholar

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🌐 Web of Science

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📍 JKU, Altenberger Straße 69, 4040 Linz

🏛 Universitas Potensi Utama, Medan, Indonesia

✂ RESEARCH PROFILE

I measure how students learn mathematics, and design the cultural and digital contexts in which they learn it.

My work pairs GeoGebra applets with AI assistance in learning trajectories, embeds local culture into statistics and numeracy through the Ethno-Flipped Classroom model, and uses Rasch Rating Scale and Many-Facet Rasch models to build and check the instruments that measure this learning.

✂ APPOINTMENTS

Jun 2026 – Nov 2026 JESH Postdoctoral Fellow / Visiting Scholar
Johannes Kepler University Linz, Austria

- Postdoctoral research on technology-enhanced mathematics education, focused on GeoGebra and AI integration.
- Design and validation of learning trajectories that support numeracy development.
- Joint projects and publications with international researchers; seminars and grant proposals.

Jun 2025 – present Associate Professor
Universitas Potensi Utama, Medan, Indonesia

- Undergraduate and postgraduate courses in mathematics, statistics, and research methodology.
- Research projects in ethnomathematics, numeracy, educational technology, and Rasch measurement.
- Supervision of research projects, theses, and publications; competitive grants and collaborations.
- Curriculum development, quality assurance, accreditation, and teacher professional development.

2019 – 2025 Senior Lecturer / Assistant Professor
Universitas Potensi Utama, Medan, Indonesia

2015 – 2019

Lecturer

Universitas Potensi Utama, Medan, Indonesia



EDUCATION

2021 – 2024

Doctorate in Mathematics Education

Universitas Negeri Medan (State University of Medan), Indonesia EQF level 8

2013 – 2015

Master's degree in Mathematics Education

Universitas Negeri Medan (State University of Medan), Indonesia EQF level 7

2012

Bachelor's degree in Mathematics Education

Institut Agama Islam Negeri Sumatera Utara (State Islamic Institute of North Sumatra), Indonesia EQF level 6



MEMBERSHIPS, METHODS, AND LANGUAGES

Memberships

Indonesian Educators Mathematics Society (I-MES), Regional Chair, North Sumatra, 2024–2026.

Psychology of Mathematics Education (PME), member, 2025–2026.

Methods

Rasch analysis (WINSTEPS, FACETS, ACER ConQuest 5), structural equation modelling, SAS, Microsoft Excel.

Languages

Bahasa Indonesia (native), English (B2).



RESEARCH GRANTS AND FELLOWSHIPS

Jun 2026 – Nov 2026

JESH Postdoctoral Fellowship

PRINCIPAL RESEARCHER

Joint Excellence in Science and Humanities, Austrian Academy of Sciences (OeAW). Host: Johannes Kepler University Linz. GeoGebra and AI integration in mathematics education; learning trajectories, research instruments, and Rasch-based analysis of numeracy learning.

Aug 2025 – Dec 2027

RIIM Competition, Wave 7: Neurodiversity-based inclusive mathematics and science education

RESEARCH TEAM MEMBER

National Research and Innovation Agency (BRIN), Indonesia. With Universitas Pendidikan Indonesia and Universitas Tanjungpura. Model development and evaluation for pre-service teacher preparation; instruments and data analysis.

Apr 2025 – Dec 2025

BIMA-DPPM Fundamental Research Grant

PRINCIPAL INVESTIGATOR

Kemdiktisaintek, Indonesia. With BRIN, JKU Linz, University of the Philippines, and Universitas Ahmad Dahlan. Deep learning trajectories and the Ethno-Flipped Classroom integrated with GeoGebra-AI for numeracy.

Apr 2024 – Dec 2025

BIMA-DPPM Domestic Collaborative Research Grant

PRINCIPAL INVESTIGATOR

Kemdiktisaintek, Indonesia. Same partners. Differentiated digital learning based on local culture and the Ethno-Flipped Classroom model for student numeracy.

2026

01

Deep learning trajectories for statistical graphs in mathematics learning: Integration of SOLO taxonomy and ethnomathematics using GeoGebra

Ramadhani, R., Soeharto, S., Prahmana, R. C. I., Bautista, G., Rahmad, I. F., Rifai, M. Z., Arifiyanti, F., & Lavicza, Z.

Discover Education.

Q2 · First author · doi:10.1007/s44217-026-01711-7

02

Cultural heritage as a mathematical context: Interpretation of set theory in spatial layout of Prambanan temple

With Universitas Ahmad Dahlan and Universitas Sriwijaya.

Infinity Journal.

Q1 · Co-author · Article page

2025

03

Assessing quality and biases in ethnomathematics-based numeracy worksheets: A Many-Facet Rasch Model analysis

Ramadhani, R., Soeharto, S., Arifiyanti, F., Prahmana, R. C. I., Saleh, A., & Lavicza, Z.

Social Sciences & Humanities Open, 12, 101736.

Q1 · First author · doi:10.1016/j.ssaho.2025.101736

04

Measuring changes in students' informal statistical reasoning skills through the ethno-flipped classroom model: Stacking and racking analysis

Ramadhani, R., Soeharto, S., Arifiyanti, F., Lavicza, Z., Syahputra, E., & Simamora, E.

Infinity Journal, 14(4), 949–972.

First author · doi:10.22460/infinity.v14i4.p949-972

05

Enhancing students' informal statistical reasoning skills through the development of a worksheet rooted in the Nias culture

With Universitas Negeri Medan.

Indexed in ERIC (EJ1474266).

Q2 · First and corresponding author · ERIC record

06

Javanese folklore with moral values: An impactful context in learning relations and functions

With Universitas Ahmad Dahlan and Universitas Sriwijaya.

Journal on Mathematics Education.

Q1 · Co-author · Article page

2024

07

The construct validity of self-regulated learning questionnaire for senior high school students: A Rasch model analysis

Ramadhani, R., Syahputra, E., & Simamora, E.

International Journal of Evaluation and Research in Education, 13(3), 2032–2038.

Q2 · First and corresponding author · doi:10.11591/ijere.v13i3.26816

08

Integrating traditional food and technology in statistical learning: A learning trajectory

With Universitas Ahmad Dahlan and Universitas Samudra.

Journal on Mathematics Education, 15(4), 1277–1310.

Q1 · First author · doi:10.22342/jme.v15i4.pp1277-1310

2023

09

Merging flipped classroom model with the ethnomathematics approach: A new flexibility learning model

With Universitas Negeri Medan.

FWU Journal of Social Sciences.

Q1 · First and corresponding author · Article page

10 Expert judgement of collaborative cloud classroom quality and its criteria using the many-facets Rasch model
With Universitas Negeri Medan and the University of Szeged.
Heliyon.

Q1 · First and corresponding author · Article page

11 Practical and social interaction justice of online learning during the COVID-19 pandemic
With Universitas Negeri Medan and Universitas Sumatera Utara.
International Journal of Evaluation and Research in Education.

Q2 · First and corresponding author · Article page

2022

12 Flipped classroom assisted Autograph in calculus learning for engineering students: A Rasch measurement study

Ramadhani, R., Bina, N. S., & Syahputra, E.

Mathematics Teaching-Research Journal, 14(4), 36–55.

First author

13 Exploration of students' statistical reasoning ability in the context of ethnomathematics: A study of Rasch model

Ramadhani, R., Saragih, S., & Napitupulu, E. E.

Mathematics Teaching-Research Journal, 14(1), 138–168.

First author

Full list on Google Scholar and Scopus.

✕ BOOKS AND CHAPTERS

2024 Students' self-regulated learning in mathematics learning
Chapter in *Application of Rasch Rating Scale Model*, pp. 6–15. Mind Metric Solution, Selangor, Malaysia.

2021 Statistics in educational research: Mathematical analysis and SPSS applications
Ramadhani, R., & Bina, N. S. Kencana, Jakarta.

✕ PATENTS AND INTELLECTUAL PROPERTY

2025	GEASTRA: GeoGebra-Supported AI for Statistical Graph Learning Copyright, e-book (granted)	EC002025175169
2025	eSTEAM-AI: cross-subject STEAM project learning app integrated with AI Copyright, computer program (granted)	EC002025133117
2021	E-School Management System method Simple patent (registered)	S00202110932
2020	Hoax-checking application Copyright, computer program (granted)	EC00202029705
2020	Statistics e-module Copyright, e-module (granted)	EC00202029706
2019	Smart Check Hoax method Simple patent (registered)	S00202007434

✕ INVITED TALKS

2026	Focus Group Discussion: OBE curriculum review Mathematics Education Study Programme, FKIP UISU
2025	Scientific article writing for Scopus journals Teacher Professional Education Study Programme, FKIP Universitas Asahan
2024	Visiting lecturer, local ethnomathematics UIN Sumatera Utara

2024	Winning research grants: Tips and tricks from successful lecturers	OZAMI Inti Sinergi
2024	Doctoral scholarships and developing an academic career	Dunia Dosen (webinar)
2024	Technical guidance for school principals: Data-based planning and assessment	Ministry of Education, Culture, Research and Technology, Balai Besar Guru Penggerak North Sumatra

Let's talk about *measures* and *motifs*.

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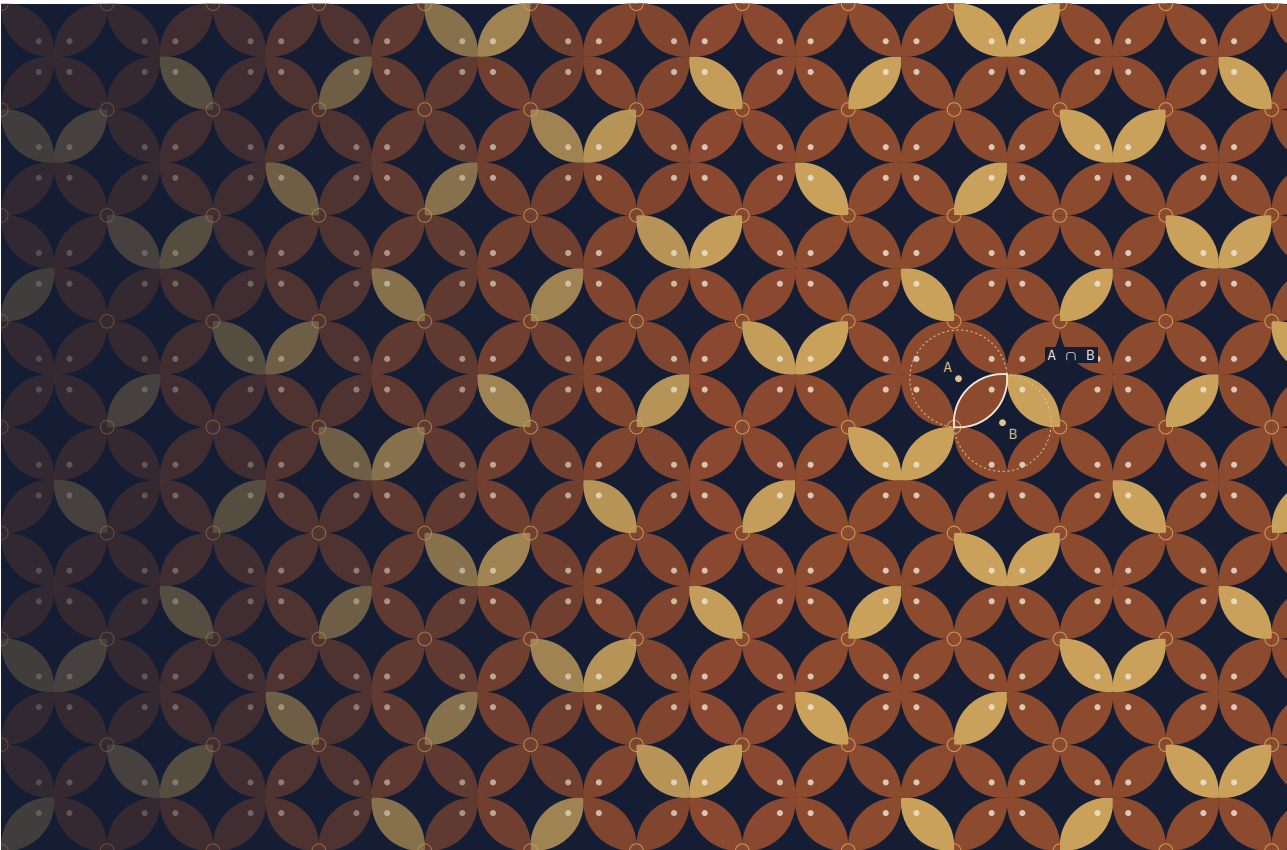


PLATE II KAWUNG, CONSTRUCTED Every petal of the Javanese kawung motif is the intersection of two circles, $A \cap B$: the set-theoretic reading of cultural form that runs through her work on ethnomathematics.